

Overview of the Transforms

Signals and Linear Systems

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- **Before Fourier:** an argument about a violin string, and the tides
- **Fourier and the Heat Equation:** the idea that was rejected for fifteen years
- **From Series to Transform:** whose name should be on it, and what the argument accidentally built
- **The Fast Fourier Transform:** Gauss, the Cold War, and five printed pages
- **The Laplace Transform:** a deaf telegraph clerk and a borrowed name
- **The z-Transform:** radar, sampled data, and where it all ends up

Nothing here is examinable

Today is the map, not the territory. Every name and date exists because somebody needed to solve a problem they could not otherwise solve — and the tool they built is the one you will spend the rest of this course learning.

One Question, Two Hundred Years



MP3, AAC



JPEG, HEIF



WiFi, 4G, 5G



MRI



LIGO, EHT

- Every one of these rests on a single question: what frequencies is this made of?
- Nobody set out to invent it. It fell out of people trying to cool a cannon, predict a tide, tame a telegraph line, and catch a nuclear test.

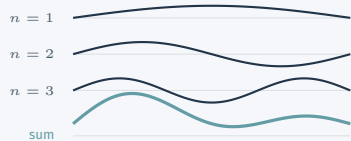
The plot of today's lecture

Watch for a pattern: **three separate times**, the right idea was rejected by the best mathematician in the room. Three times, the engineers used it anyway.

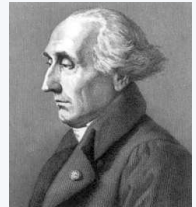
Before Fourier: An Argument About a String

1747–1759: A Fight Over a Violin String

- **1747:** d'Alembert writes down the wave equation for a plucked string. **1748:** Euler finds its normal modes — the pure shapes that vibrate without changing shape.
- **1753, Daniel Bernoulli** makes the leap: **every** possible motion of the string is a sum of those modes. He argues it from physics, and offers no proof.
- **1759, Lagrange** demolishes him. A sum of smooth sines cannot reproduce an arbitrary shape — a string plucked to a sharp corner, say. Euler agrees. Bernoulli is buried.
- He was right. It took another seventy years to show it.



Daniel Bernoulli
“always a sum of modes”



Joseph-Louis Lagrange
“it cannot be”

Meanwhile, a Second Family Tree Is Growing

- Long before anyone said “transform”, mathematicians were packing a sequence or a function into one object so calculus would turn into algebra.

de Moivre, 1730: generating functions

$$A(z) = \sum_{n=0}^{\infty} a_n z^n$$

Flip the sign of the exponent and you have written the **z-transform**, 217 years early.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Euler, 1744: an integral that does the same job

$$\int_0^{\infty} e^{-sx} f(x) dx$$

From 1785 **Laplace** uses exactly this to turn difference and differential equations into algebra – while also writing the theory of probability and a mechanics of the solar system.

Same trick, two costumes

A sum over z^{-n} for sequences; an integral over e^{-st} for functions. Later you will prove they are the same object, joined by $z = e^{sT}$.

Laplace, the Tides, and a Machine Made of Brass

- **1775–76:** Laplace replaces the old “the Moon pulls a bulge of water” picture with his tidal equations — the ocean as a driven, rotating dynamical system.
- The engineering consequence: a tide is a **sum of sinusoids** whose frequencies come from astronomy and whose amplitudes and phases come from the harbour.
- **1872:** William Thomson (Lord Kelvin) builds a machine of pulleys and wire that adds ten of those sinusoids and draws the tide — a year of predictions in about four hours of cranking.
- **October 1943:** a handwritten letter marked most urgent reaches Arthur Doodson in Liverpool, asking for 1944 tides at a place called only “**Position Z**”. He gets harmonic constants and no map, and runs them on Kelvin’s 1872 machine.
- Position Z was Normandy. Those numbers picked the hour of D-Day.



Pierre-Simon Laplace



Kelvin's tide-predicting machine,
1872

Analysis and synthesis, in brass

The first Fourier synthesiser was made of gears — and people were adding up sinusoids to predict the world long before anyone proved they were allowed to.

Fourier and the Heat Equation

Joseph Fourier (1768–1830): Not Only a Mathematician

- Son of a tailor, ninth of twelve children, orphaned by ten. He had finished Bézout's six-volume mathematics course by fourteen.
- Trained for the priesthood but never took vows. Joined the Revolution instead, was arrested in July 1794 and expected the guillotine — Robespierre was executed first.
- **1798**: sailed with Napoleon to Egypt as scientific adviser, helped found the Cairo Institute and edited the 24-volume *Description de l'Égypte*.
- As Prefect of Isère he drained the Bourgoin marshes and built the road to Turin by day, and invented Fourier analysis at night.
- In Grenoble he took an interest in a schoolboy named **Jean-François Champollion** and showed him his Egyptian collection. Champollion deciphered the Rosetta Stone in 1822.



Joseph Fourier

The eccentric

Convinced heat was healthy, he kept his rooms hotter than Hades and went about wrapped in coats in summer.

The Actual Problem: How Does Heat Spread?

- Fourier's question was industrial and imperial: how does temperature evolve inside a solid — a cannon, a mine shaft, the Earth?

- His answer was the **heat equation**:

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

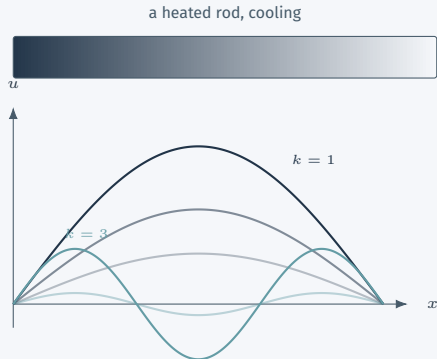
- Easy to write, hard to solve, because $u(x, 0)$ can be any shape.

The one shape that behaves

If $u(x, 0) = \sin(kx)$ then for all later times

$$u(x, t) = e^{-\alpha k^2 t} \sin(kx).$$

A sinusoid **diffuses into itself**. It only shrinks — and the faster it wiggles, the faster it dies.



High frequencies vanish first: that is why a hot object forgets its detail before it forgets its warmth.

- If the only shape Fourier can solve is a sinusoid, then he must build every shape out of sinusoids. So he simply announces that he can:

Fourier's claim, 1807

$$f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx), \quad a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx \, dx$$

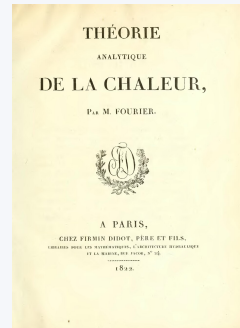
Any periodic function. Including ones with corners. Including ones with jumps. Including ones nobody has written a formula for.

- Then the whole problem dissolves: split the initial temperature into sinusoids, shrink each one by its own $e^{-\alpha k^2 t}$, and add them back up.
- Notice what happened. A hard problem in x became an easy problem in k . **That move — change domain, solve trivially, change back — is the entire subject of this course.**

The transform pattern

hard operation in one domain $\rightarrow$ **multiplication** in the other $\rightarrow$ come back

- Fourier reads On the Propagation of Heat in Solid Bodies to the Institut de France. The referees are **Lagrange, Laplace, Monge and Lacroix**.
- Lagrange is the greatest living analyst — and the same man who demolished Bernoulli's sum-of-modes claim 48 years earlier. He objects again. Publication is blocked.
- The Institut turns the question into the **1811 prize essay**. Fourier enters and wins in January 1812 — with a public censure attached: “the manner in which the author arrives at these equations is not exempt of difficulties”. Still no printing.
- So he waits. In **1822** he publishes it himself as *Théorie analytique de la chaleur*. He is by then Secretary of the Académie, and nobody can block him any more.

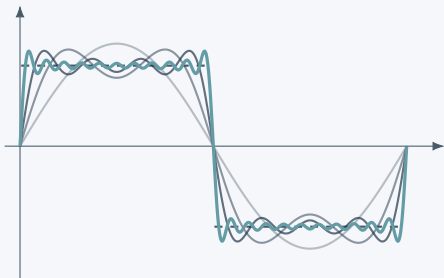


Paris, 1822 — fifteen years after the memoir

Worth remembering when your idea gets rejected

Lagrange was not being stupid. He was being careful, and on the narrow mathematical point he was even partly right. He was still wrong about the thing that mattered.

What He Was Claiming, in One Picture



1, 2, 4 and 11 sine terms against a square wave

- A square wave has **corners and jumps**. Sines have neither.
- Yet with enough of them, at exactly the right amplitudes, the sum flattens onto the square almost everywhere.
- This is the picture Lagrange refused to believe.

Keep an eye on the jump

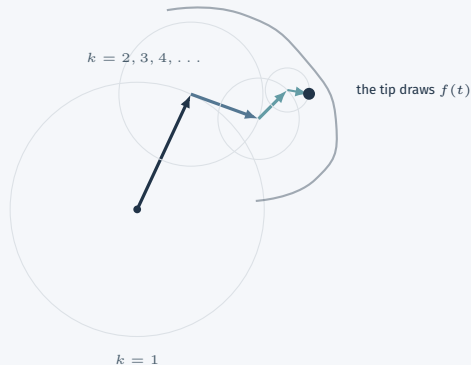
Look at the little spike that refuses to lie down at the discontinuity. It has its own story, and the wrong man got the credit. We will come back to it.

The Same Idea, as Rotating Arrows

- Write the series in complex form and every term becomes a **vector spinning at a constant rate**:

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

- $|c_k|$ is the length of the k -th arrow, $\angle c_k$ is where it starts, k is how fast — and which way — it turns.
- Attach each arrow to the tip of the one before it. The final tip traces $f(t)$.
- With enough little arrows you can draw a square wave, a heartbeat, or your own signature.



Worth 25 minutes of your evening

3Blue1Brown, But what is a Fourier series? From heat flow to circle drawings: youtu.be/r6sGWTCMz2k

Fourier's Other Discovery

- Having built a theory of heat, Fourier pointed it at the largest solid body within reach: the Earth.
- In papers of **1824** and **1827** he worked out how warm a planet of Earth's size, at Earth's distance from the Sun, ought to be — and found it should be far colder than it is.
- His explanation: the atmosphere lets sunlight in more easily than it lets heat back out. That is the **greenhouse effect**, described in 1824. He compared it to the glass of a hothouse — which is where the name came from — and every climate model today is a descendant of it.

One person, two fields

The man who gave signal processing its central tool also founded climate science. Both fell out of the same stubborn question about heat.



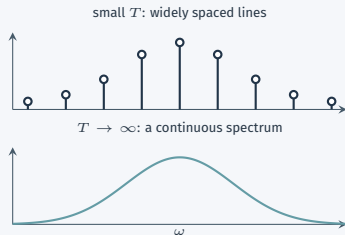
“Heat, like gravity, penetrates every substance of the universe.”

— *Théorie analytique de la chaleur*, 1822

From Series to Transform — and Whose Name Goes On It

From a Ring to a Line

- A Fourier series answers a question about a periodic signal: it lives on a circle, so only whole-numbered harmonics fit and the spectrum is a set of **spikes** at $k\omega_0$.
- Now stretch the period. As $T \rightarrow \infty$ the spacing $\omega_0 = 2\pi/T$ shrinks to zero, the spikes crowd together, and the sum becomes an integral.



The continuous-time Fourier transform

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

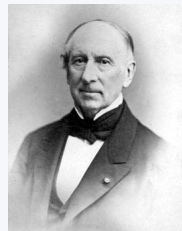
Same idea, one fewer restriction: the signal no longer has to repeat.

Why this matters more than it looks

A series describes one signal. A transform is an **operation** you can apply to anything — and that is the object the rest of the story is about.

So Whose Transform Is It?

- The integral pair was in the air, and at least four people had a claim.
- **Laplace, 1809** gives an integral formula in the Journal de l'École Polytechnique. Fourier's own 1822 book credits him for it: "Elle a été donnée par M. Laplace ...nous ne faisons que l'appliquer" — it was given by Mr Laplace; we merely apply it.
- **Fourier, 1811–1822** writes the forward and inverse sine and cosine transforms explicitly, and finds that differentiating n times multiplies the transform by $(j\omega)^n$.
- **Cauchy, 1815** and **Poisson, 1816** publish the same formula independently — and, unlike Fourier, prove that the integral converges.



Cauchy, 1815



Poisson, 1816

Fourier spent his last years fighting priority claims from Biot and Poisson.

The irony

The man who blocked Fourier in 1807 is credited by Fourier in 1822 — and the rigour Lagrange demanded came from Cauchy and Poisson, who were racing Fourier for the result.

Nobody Called It “The Fourier Transform” Until 1915



- Plancherel’s sentence, in 1915, is where the object gets its modern name: “Nous nommerons $F(x)$ la transformée de $f(x)$.”
- So: invented around 1809–1816, named after one of its four discoverers in 1915, final form in 1934 — a hundred and twenty years after it was first written down.

The Argument Nobody Could Settle — and What It Built

- Lagrange's objection would not die, because “does this series actually converge?” turned out to be genuinely hard. Answering it rebuilt mathematics.
- **1829, Dirichlet** gives the first honest convergence conditions, and shows that at a jump the series converges to the midpoint.
- **1854, Riemann** needs a better notion of “area under a curve” for trigonometric series — and invents the **Riemann integral**.
- **1870, Cantor** asks a small question: can a function have two different trigonometric series? Chasing the exceptional sets where his proof failed, he invents **set theory**.
- **1902, Lebesgue** builds a new integral, again with Fourier series in mind. **1966, Carleson** finally settles the question — 207 years after Lagrange raised it.



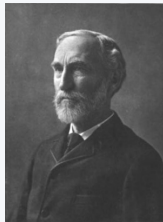
Georg Cantor

Scale of the accident

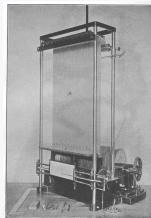
Infinity, cardinality, the real line, measure theory: all downstream of a row about a vibrating string.

The Overshoot That Would Not Go Away

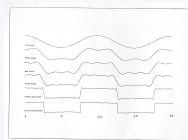
- **1848:** an English mathematician, **Henry Wilbraham**, notices the spike at a jump, analyses it correctly, publishes — and is comprehensively ignored for the next 66 years.
- **1898:** **Albert Michelson** — of Michelson–Morley — builds an 80-coefficient harmonic analyser out of springs and gears. Fed a square wave, it keeps drawing that same stubborn ear at every jump.
- He writes to Nature assuming his machine is broken. A public argument with A. E. H. Love follows in the same pages.
- **1899:** **J. W. Gibbs** settles it. The machine is fine. The overshoot is real, it is about 9% of the jump, and it never goes away — it only gets narrower.
- **1906:** Bôcher names it the **Gibbs phenomenon**, after the man who explained it second.



J. Willard Gibbs



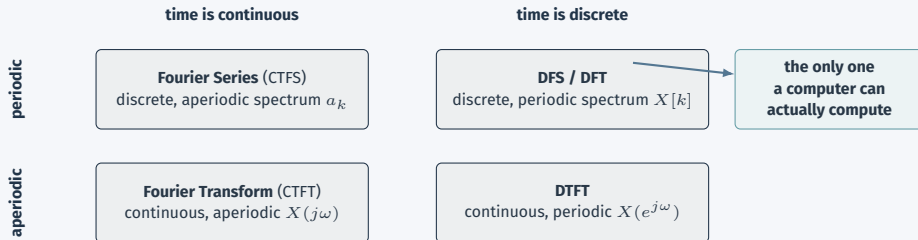
Michelson's harmonic analyser,
1898



His printed approximations — the ears are visible at every
jump

Four Fourier Transforms, One Idea

- Fourier's series needs a periodic, continuous signal. Let the period run to infinity and the sum becomes an integral. Sample the signal and the frequency axis becomes periodic. Do both and everything becomes finite — which is the only case a computer can touch.



The rule that generates the whole table

Discrete in one domain $\iff$ periodic in the other. Next class, we derive every arrow in this square.

The Fast Fourier Transform

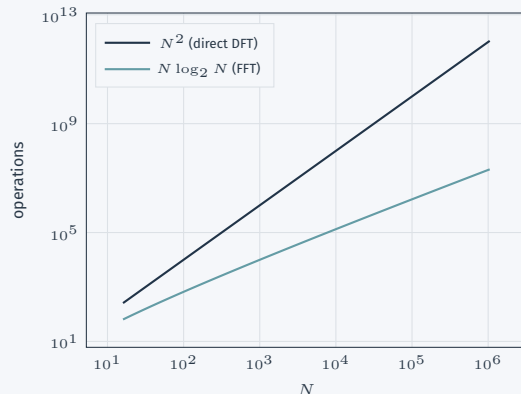
The DFT Is Correct, and Uselessly Slow

- Computing $X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}$ straight from the definition costs N multiplications for each of N values of k : $O(N^2)$.
- Double the data and you quadruple the work. That is the whole problem.

N	N^2	$N \log_2 N$
1 024	10^6	10^4
10^6	10^{12}	2×10^7
10^9	10^{18}	3×10^{10}

The question of the next few classes

Nothing about the DFT is wrong. It is simply being computed stupidly — and the fix had been sitting in a notebook since 1805.



1805: Gauss Gets There First, and Tells Nobody

- Gauss is chasing the newly discovered asteroids **Pallas** and **Juno**. He has a handful of noisy sightings and needs to interpolate a periodic orbit through them.
- Doing the trigonometric coefficients by hand is brutal, so he finds a shortcut: split the samples into two interleaved halves, transform each, then combine. Then do it again. And again.
- That is the radix-2 decimation-in-time FFT — the identical algorithm you will derive later in this course.
- He writes it in a private notebook, in Latin, in a notation of his own. He never publishes it. It surfaces only in 1866, in his posthumous collected works, where essentially nobody reads it.



Carl Friedrich Gauss

Two years before Fourier

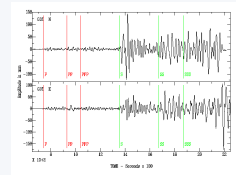
The fast Fourier transform was written down before the Fourier transform was — and then lost for 160 years.

Why Anyone Cared in 1963

- After 1945 the United States offered to give up its arsenal if others would forgo theirs. The offer failed, and the arms race started.
- By the early 1960s both sides wanted a test ban, and both had the same obstacle: **verification**.
- The **Partial Test Ban Treaty** of 1963 outlawed testing in the atmosphere, in space and underwater — everything visible from outside. It could not outlaw **underground** testing, because nobody could prove a violation.
- An underground test shows up only as a shiver in the ground, recorded thousands of kilometres away, buried in a planet that shivers anyway.



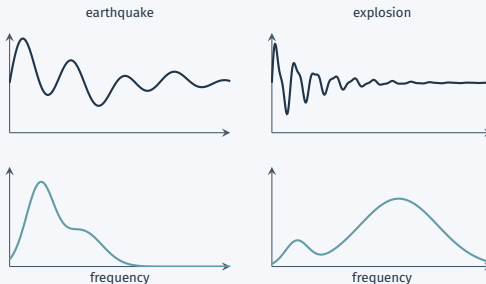
Nevada Test Site, 1957 — the kind of test a treaty could police



A seismic record: everything looks like everything else

The Signature Is in the Spectrum

- An earthquake is a slip along a fault: it releases its energy over seconds, so the energy sits at **low** frequencies.
- A bomb is a point source going off at once: sharp onset, far more **high**-frequency content.
- In the time domain the two traces look like each other. In the frequency domain they do not.
- So: ring the Soviet Union with seismometers and take the DFT of everything they record, continuously, forever.
- At $O(N^2)$, “continuously, forever” is not a plan. It is a wish.



A Meeting, and a Cover Story

- **1963.** At a meeting of President Kennedy's Science Advisory Committee on detecting underground tests, the Princeton statistician **John Tukey** mentions that he has been working on a way to speed up the Fourier transform.
- Also in the room is **Richard Garwin** of IBM — a physicist who, at 23, had done the design work on the first hydrogen bomb. He grasps the consequence instantly.
- Back at IBM Yorktown Heights, Garwin asks for a mathematical analyst and is given **James Cooley**. By Garwin's own later account, he did not tell Cooley the real motive: the stated application was studying the spins in a crystal of helium-3.
- Tukey also gave us two words you use every day: bit — Shannon credits him for it in 1948 — and, in 1958, the first printed use of software.



John W. Tukey



James W. Cooley



Richard L. Garwin

J. W. Cooley and J. W. Tukey

An Algorithm for the Machine Calculation of Complex Fourier Series
Mathematics of Computation **19** (1965), pp. 297–301.

- $O(N^2) \rightarrow O(N \log_2 N)$. At $N = 2^{20}$ that is a factor of roughly **50 000**.
- The 1964 demonstration ran about a hundred times faster than the existing code, and could keep up with the seismic data in real time.
- A calculation that took a day now takes under two seconds. Entire fields that were theoretically possible became routine overnight.

How the profession rates it

The FFT is on the IEEE/AIP list of the **Top 10 Algorithms of the 20th Century**, and the 1964 demonstration on an IBM 7094 carries an **IEEE Milestone** plaque.

Gilbert Strang called it “the most important numerical algorithm of our lifetime”.

- Several classes of this course are, in effect, a careful reading of those five pages and of what came after them.

The Part That Stings

- The algorithm worked. The monitoring network it made possible got built. But by 1965 the moment had gone: the United Kingdom, France and China had all joined the club, and the political will for a comprehensive ban went with them.
- The **Comprehensive Nuclear-Test-Ban Treaty** was not opened for signature until **1996** — and it has still not entered into force.
- It is watched anyway. The CTBTO's International Monitoring System is **321 stations and 16 laboratories** across 89 countries, **170 of them seismic**, running Fourier transforms day and night — exactly the machine Garwin wanted, thirty years late.

The counterfactual

Veritasium's claim — and it is a claim, not a proof — is that an FFT a decade earlier might have made a comprehensive test ban verifiable while it was still politically possible. Either way, one thing is certain: **the algorithm arrived too late for the treaty, and exactly in time for everything else.**

The Laplace Transform

Fourier

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Needs the integral to converge. A growing signal such as $e^{2t}u(t)$ simply has no Fourier transform — and neither does any unstable system.

Laplace

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt, \quad s = \sigma + j\omega$$

Multiply by $e^{-\sigma t}$ first, then take the Fourier transform. Pick σ large enough and the integral converges after all.

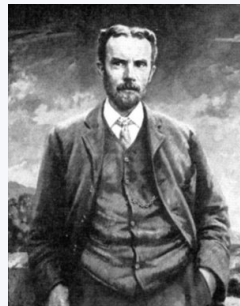
- The price of that freedom is that you have to say which σ you used. That set of values is the **region of convergence**, and it will take up a good part of the second half of this course.

The one-line summary of the whole course

Fourier lives on a **line** ($j\omega$). Laplace lives on a **plane** ($\sigma + j\omega$). Everything else is bookkeeping.

Oliver Heaviside (1850–1925): The Outsider

- Scarlet fever left him partly deaf as a child. He left school at sixteen, taught himself Morse, electricity, Danish and German, and worked as a telegraph clerk until the deafness ended that too. Then he simply stopped having a job, and did physics alone in a back room for decades.
- He reduced Maxwell's **twenty equations in twenty variables** to the **four** every physics student now learns, and coined impedance, inductance and reactance.
- And he invented **operational calculus**: write p for d/dt , treat it as ordinary algebra, and solve circuit equations as if they were fractions. It worked; telegraph engineers used it daily. It was also, by 1890s standards, entirely unjustified — and the Royal Society rejected the paper for “irredeemable inadequacies in proof”.
- He was right about the sky too: in 1902 he predicted a conducting layer in the upper atmosphere that reflects radio waves. Confirmed in 1923, it is still called the **Heaviside layer**.



Oliver Heaviside

His reply to his critics

“Shall I refuse my dinner because I do not fully understand the process of digestion?”

Vindication — and a Borrowed Name



- Bromwich's inversion integral is the one you will meet later this term: $x(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st} ds$.

The name is an accident of history

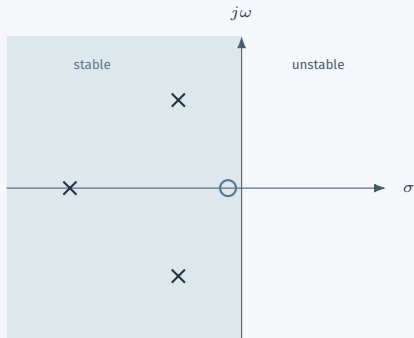
Laplace never wrote “the Laplace transform”. Heaviside, who made it an engineering tool, did not get his name on it either. The label settled in the 1920s and 30s — roughly **150 years** after Laplace's work, and after the one man who actually needed it was dead.

What the Laplace Transform Bought Us

- A differential equation becomes a **polynomial fraction** $H(s) = \frac{N(s)}{D(s)}$: the transfer function.
- Its **poles** tell you whether the system rings, settles or explodes — without your solving anything.
- Feedback becomes algebra: $\frac{Y}{X} = \frac{G(s)}{1 + G(s)H(s)}$.
- Out of that came root locus, Bode and Nyquist plots, the PID controller, aircraft autopilots, the Apollo guidance computer, power-grid stabilisers, and every op-amp circuit you will ever design.

Later this term

Heaviside's "just do algebra with d/dt " is now a rigorous method — and it is the one you will use to solve differential equations for the rest of your degree.



Poles (x) left of the axis: the system settles. Right of it: it runs away.

The z-Transform and the Digital Age

- Wartime tracking radar does not see continuously. It sees in **pulses**: one sample per sweep of the antenna.
- A control loop closed around such a sensor is not governed by a differential equation but by a **difference** equation — and the Laplace transform alone does not handle it cleanly.
- At the MIT Radiation Laboratory, **Witold Hurewicz** solves it in the 1947 volume Theory of Servomechanisms by reviving the old generating-function idea for sampled sequences.
- Hurewicz was a topologist by trade — the Hurewicz theorem and the Hurewicz fibration are his — who had wandered into control engineering because of the war. He died in 1956 after falling from a Mayan pyramid at Uxmal, on a conference excursion.

The pattern repeats

continuous $\rightarrow \int x(t)e^{-st} dt$

sampled $\rightarrow \sum x[n]z^{-n}$

A war-driven engineering need, solved by reaching back two hundred years to Laplace's generating functions.

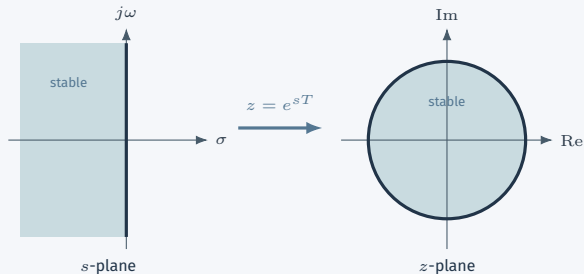
1952: The Name, and the Picture That Ties It Together

- **Ragazzini and Zadeh**, in the sampled-data group at Columbia, give it the name we still use: the **z-transform**. The coincidence with Zadeh's initial appears to be exactly that — the letter was already standard in the difference-equation literature.
- Zadeh went on to invent **fuzzy logic** in 1965, which is a second career most people would settle for.
- Sample $x(t)$ every T seconds; the Laplace transform of the sampled signal is a function of e^{sT} . Call that z :

The bridge

$$z = e^{sT}$$

the $j\omega$ axis wraps onto the **unit circle**; the stable left half-plane maps to the **inside** of it.



Lotfi A. Zadeh

We make this precise later in the course. Sampling itself — Nyquist 1928, Shannon 1949 — gets a class of its own.

Where the Transforms Actually Live Now



JPEG, HEIF

DCT — Nasir Ahmed, 1972



MP3, AAC

tuned on Tom's Diner



WiFi, 4G, 5G

the OFDM modulator is an IFFT



MRI

the image is an inverse FT



LIGO, EHT

black holes, 2015 and 2019



DSP, control

Laplace and z , everywhere

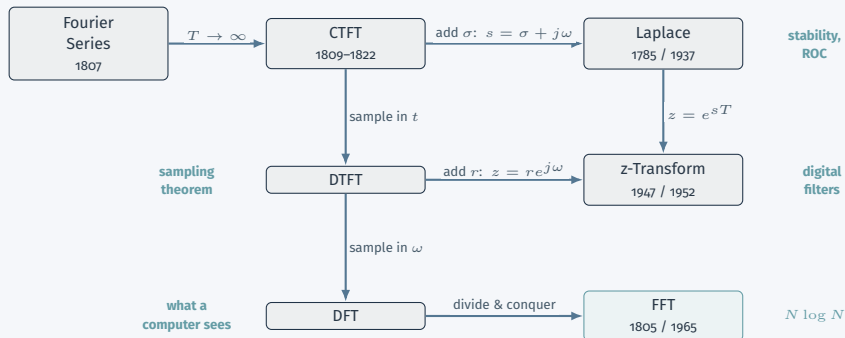
One more rejection, for symmetry

Nasir Ahmed proposed the discrete cosine transform to the US National Science Foundation in 1972. Rejected: a reviewer found it “too simple”. It is now in every JPEG on Earth.

And one for fun

Karlheinz Brandenburg tuned MP3 on Suzanne Vega's Tom's Diner: at bitrates where everything else sounded fine, her voice did not. He listened to it thousands of times.

The Family Tree



Two centuries, one semester

Every arrow on this page was a fight. A dozen people, several of whom were told they were wrong, and one who was ignored for 160 years. **You get all of it in one term.**

Watch

- 3Blue1Brown, But what is a Fourier series? youtu.be/r6sGWTCMz2k
- 3Blue1Brown, But what is the Fourier transform? youtu.be/spUNpyF58BY
- Veritasium, The Most Important Algorithm of All Time
youtu.be/nmgFG7PUHfo

Read

- Oppenheim, Willsky & Nawab, Signals and Systems, §3.1 — A Historical Perspective. Two pages; read them before next class.
- Heideman, Johnson & Burrus, Gauss and the History of the Fast Fourier Transform, IEEE ASSP Magazine, 1984.
- [IEEE Pulse](#), Highlights in the History of the Fourier Transform.

Next class

Introduction to the DTFT and the DFT. We stop telling stories and start deriving — beginning with what happens to Fourier's idea when time itself arrives in samples.